



SMC

S1 Science Course

Nourishing life



Name:

Learning Outcomes

Statement			
1. Life giving plants			
Understand the importance of plant life to the environment			
Give examples of how plants benefit humans			
2. Parts of a plant			
Identify organs found in flowering plant			
State the function of organs found in flowering plant			
Understand that starch is vital energy source in plant-based food			
Test foods for the presence of starch			
3. Photosynthesis			
The word equation for photosynthesis			
State the function of seed parts			
4. Flower power			
Identify the names and functions of structures in a flower			
5. Pollination			
Describe what happens during pollination in flowering plants			
State the function of the pollen tube			
Recognise the development of fruit from seeds			
Understand the importance of seed dispersal			
6. Plant eaters			
Identify the parts of plants that are edible			
Describe the features of consumers based on skulls and teeth			

Identify the main parts of a human tooth			
Statement			
7. Breaking down plant food			
State what is meant by digestion			
Describe the journey of food from gum to bum			
Identify the parts of the human digestive system			

Lesson 1 – Life giving plants

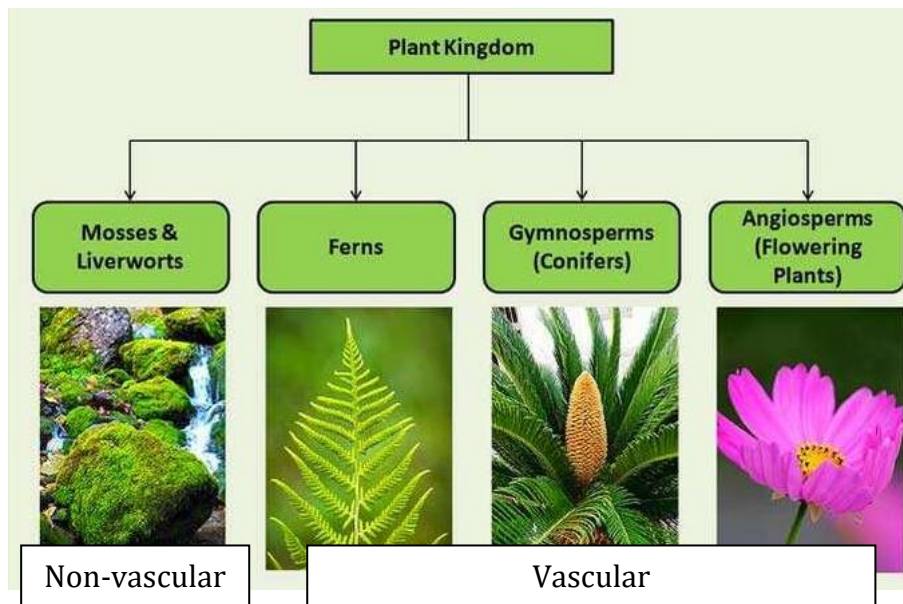
Life on Earth depends on plants.

Every single day and in every single way, humans use plants. From the **food** we eat to the **cars** we drive to the **medicines** we take and the **clothes** we wear, we would not be able to live without plants.

Plants also benefit the environment: reducing the risks of **floods** and **drought**, storing **carbon dioxide**, and adding **oxygen** to the atmosphere.

Plants are also a source of inspiration and enjoyment for many.

There are approximately 391,000 species of **vascular plants** (plants with stems) currently known to science, of which about 369,000 species (or 94 percent) are **flowering plants**.



Scientists estimate that humans can eat as many as **300,000** plant species, and yet we only consume about **200**!

Plants are the only organisms that can **make their own food**. They do this using the process of **photosynthesis**. Animals cannot make their own food, so they must **consume** plants (or other organisms that ate plants) to get food. All the food we eat can be traced back to photosynthesising plants.

Questions: Read the above text and answer the following questions:

1. Name an everyday activity for humans that requires plants?

2. Name a benefit of plants in the environment?

3. What group of plants makes up the majority of plants?

4. Of the other types of plants in the plant kingdom, which are found in water bodies?

5. What percentage of edible plant species do humans consume?

This topic will follow food: from how it is **made by plants**, through to how we break down food in our **digestive systems**, including some of our unique **adaptations** to eat food.

But first, let's look at the big picture and explore the diverse range of plants in our world.

Activity 1 In pairs you will get several fact sheets that you will add to your notes in summary form about the “world of plants”

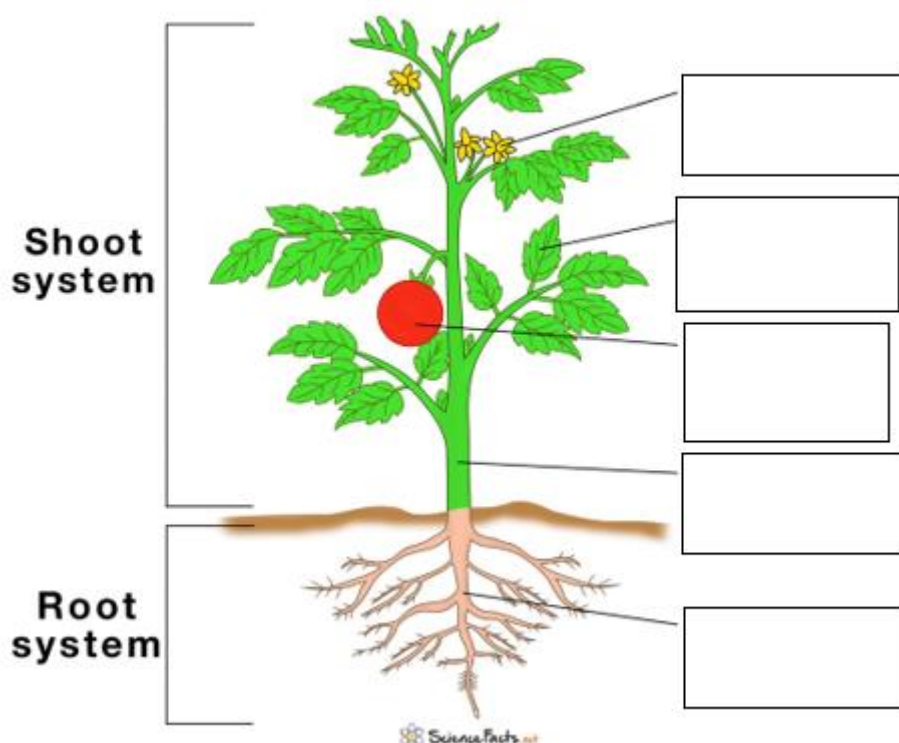
Happy Families	Evolution
Waterworld World of plants Populate this grid with all the interesting facts about plants	
Life in extremes 1) Hot and dry climates 2) Jungle	Poisons

Lesson 2 – Parts of a Plant

What makes up a plant?

There are **basic structures** that all flowering plants (94% of all plants on earth) share.

Use the diagram and images your teacher will show you and identify the key **structures** on the flowering plant. Each part of the plant has a **specific function**.



Task 1: Identify which job is associated with which part of the plant and match them up with an arrow.

Plant part		Job
Leaves		Flexible and strong, helps the plant stand up from ground and protects the tubes that carry food and water
Stem		These enable plants to make seeds and therefore more plants
Root		Thin, flat organs are where plants make their own food
Flower		These seek out and suck up water and nutrients , while acting as an anchor

Finding food in plants

Photosynthesis makes sugars in the **leaves**. These sugars are often passed to other parts of the plants and converted into starch for storage.

When starch is present, iodine solution changes from _____
to _____



Experiment 1: Collect some foods made from plants and test them using iodine. You should scratch the surface of the food so that the iodine can get into the structure.

<i>Food tested</i>	<i>Iodine colour</i>	<i>Starch present or absent</i>



Wear safety glasses
when using

1. How did you know if starch was present in the plant sample?

2. Name the process that plants use to make sugars.

3. Why does the plant convert the sugars into starch?

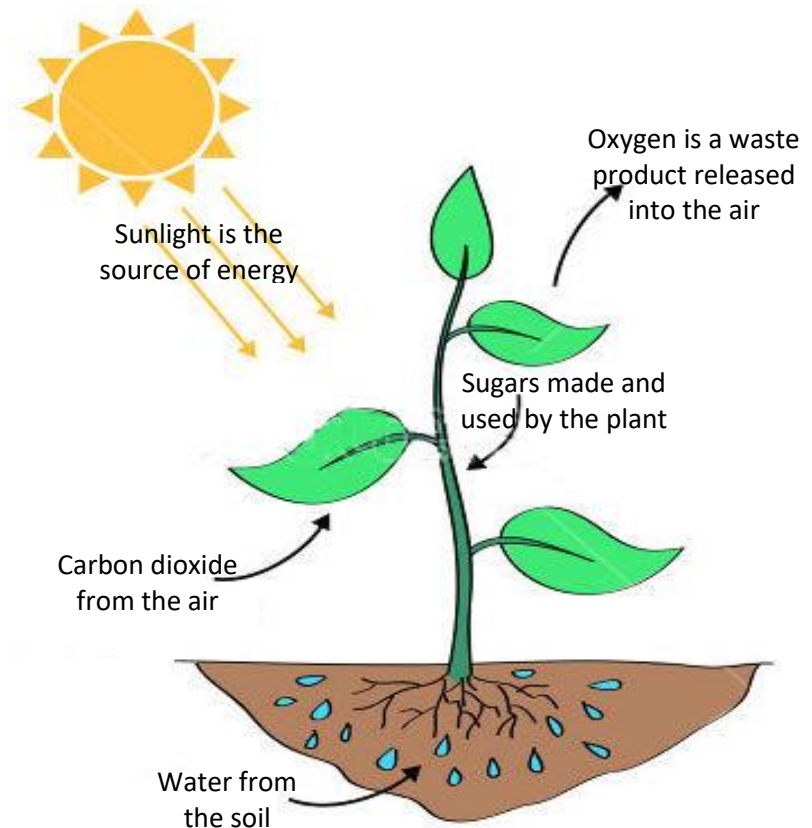
Lesson 3 – Photosynthesis

The need for light:

Plants need light as their source of **energy** so that they can do photosynthesis.

Plants have evolved to be very efficient at converting this **light energy** into **chemical energy** in sugars.

The process of photosynthesis uses simple chemicals from the plant's surroundings to produce sugars and a waste product.



Questions

1. What type of energy is needed to power the process of photosynthesis? _____
2. What are the two chemical substances that are needed for photosynthesis?
_____ and _____
3. What is the useful product of photosynthesis?

4. What is the waste product of photosynthesis and what happens to it?

5. Complete the word equation of photosynthesis.

Raw materials

Products

light energy

_____ and _____



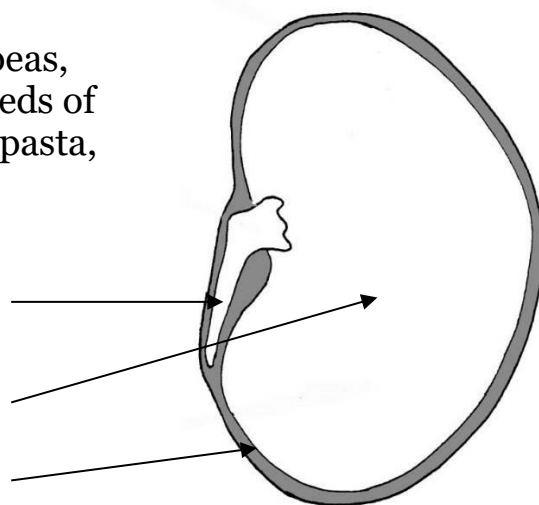
_____ and _____

Seeds

Many of the foods that we eat are seeds. Rice, peas, beans, nuts and sweetcorn are all seeds. The seeds of wheat, barley and oats are used to make bread, pasta, beer and porridge

Label the following structures:

- seed coat
- embryo plant
- food store

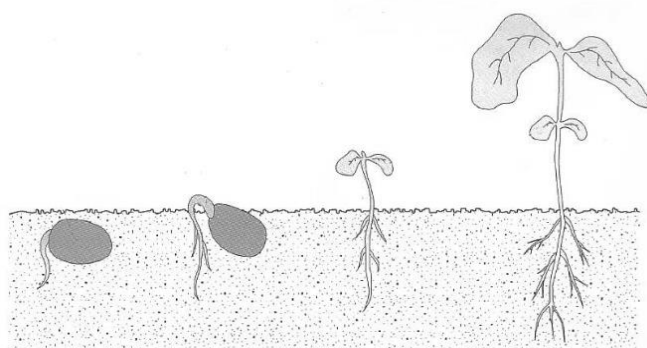


The seed can grow to form a new plant, so it has three important structures to make sure that this happens.

Describe how you could prove that the food store contains starch.

The process that happens when a seed grows to form a new plant is called **germination**.

Describe the functions of the seed structures.



1. Seed coat

2. Embryo

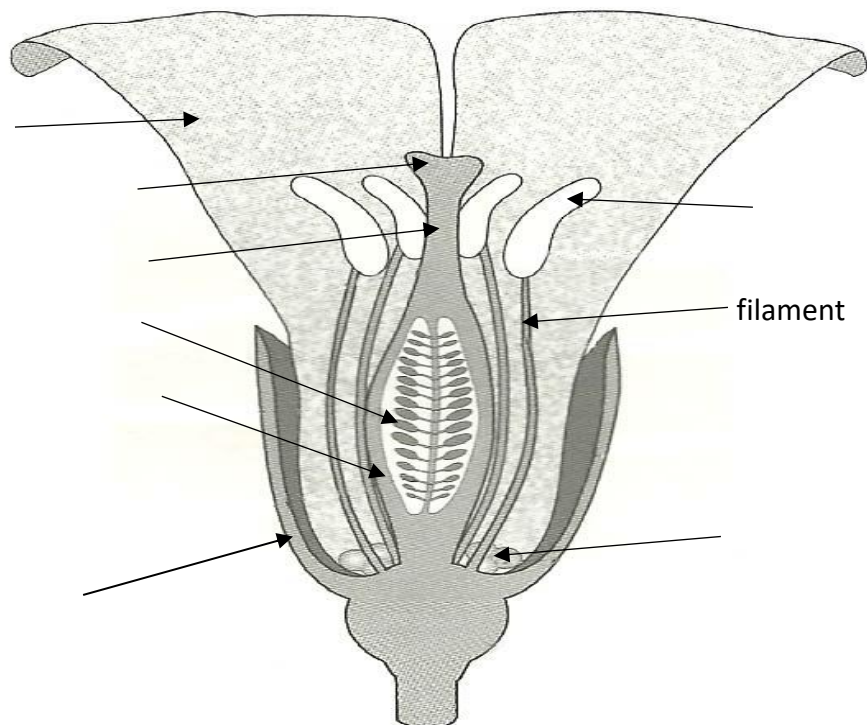
3. Food store

Lesson 4 – Flower power

In flowering plants, flowers enable plants to make **seeds** and therefore more plants.

Task 2: Label the following structures:

- Anther
- Nectary
- Ovary
- Ovule
- Petal
- Sepal
- Stigma
- Style



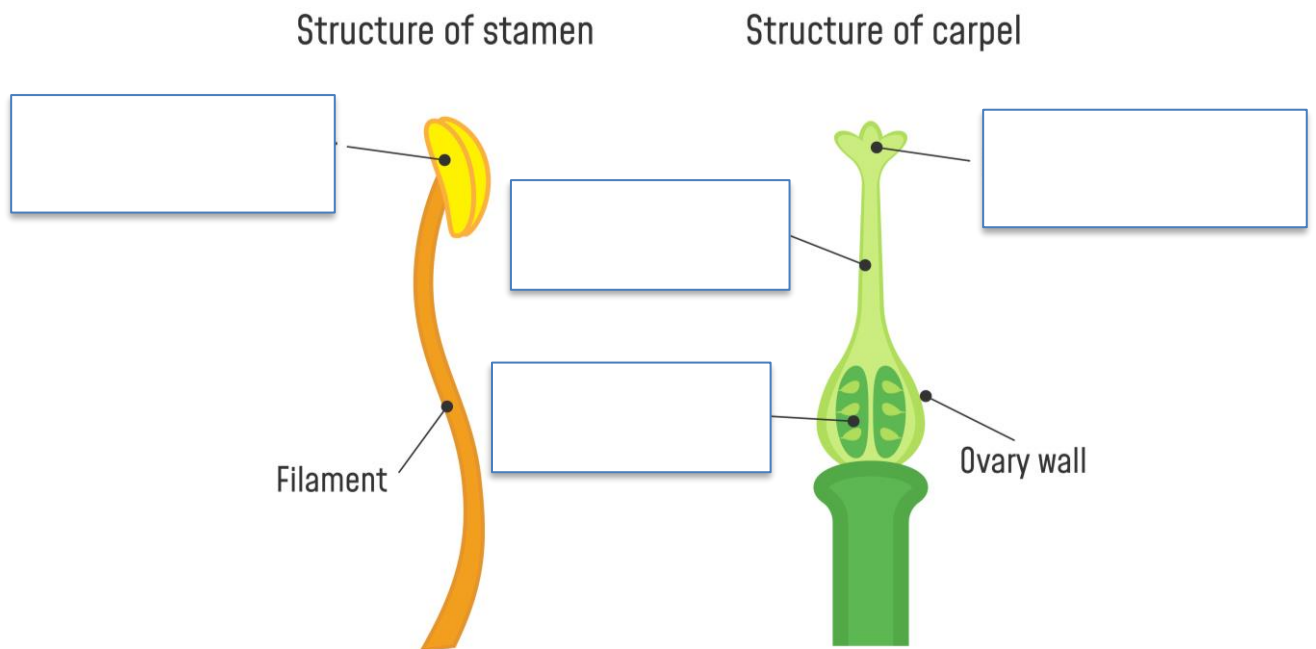
Carpel is the female part of the flower.

Which **four** structures make the carpel?

Stamen is the male part of the flower.

Which **two** structures make up the stamen?

Add labels to the male and female parts of a flower:



Activity 2: Use the card sort to complete the table by adding the missing structures and functions of an ***insect pollinated*** flower.

<i>Structure</i>	<i>Function</i>
Sepal	
	Coloured and scented to attract insects
Anther	
Stigma	
	Develops into fruit. Contains <i>ovules</i> which become seeds.
	Produces nectar to attract insects



Experiment 3: Flower dissection display

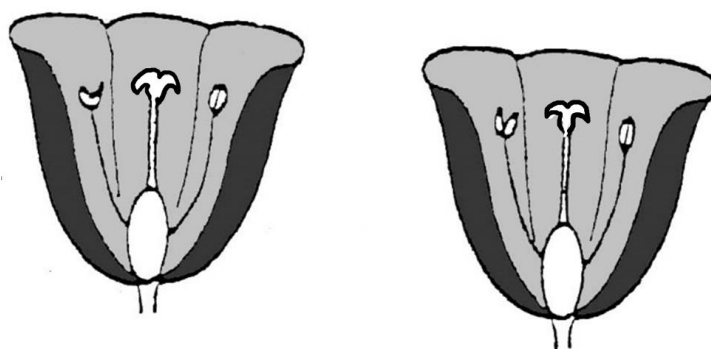
After you dissect your flower, stick the parts here with your own labels

Lesson 5 –Pollination

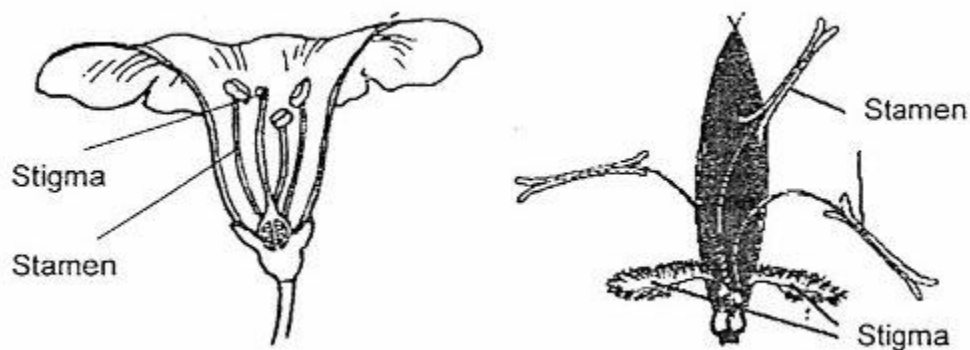
This is when pollen is moved from the _____ to a _____.

On the diagram below:

1. use red to colour the anthers on both flowers
2. use blue to colour the stigma on both flowers
3. add an **arrow** to show how one flower pollinates the other.



Comparing adaptations of wind-pollinated and insect-pollinated flowers

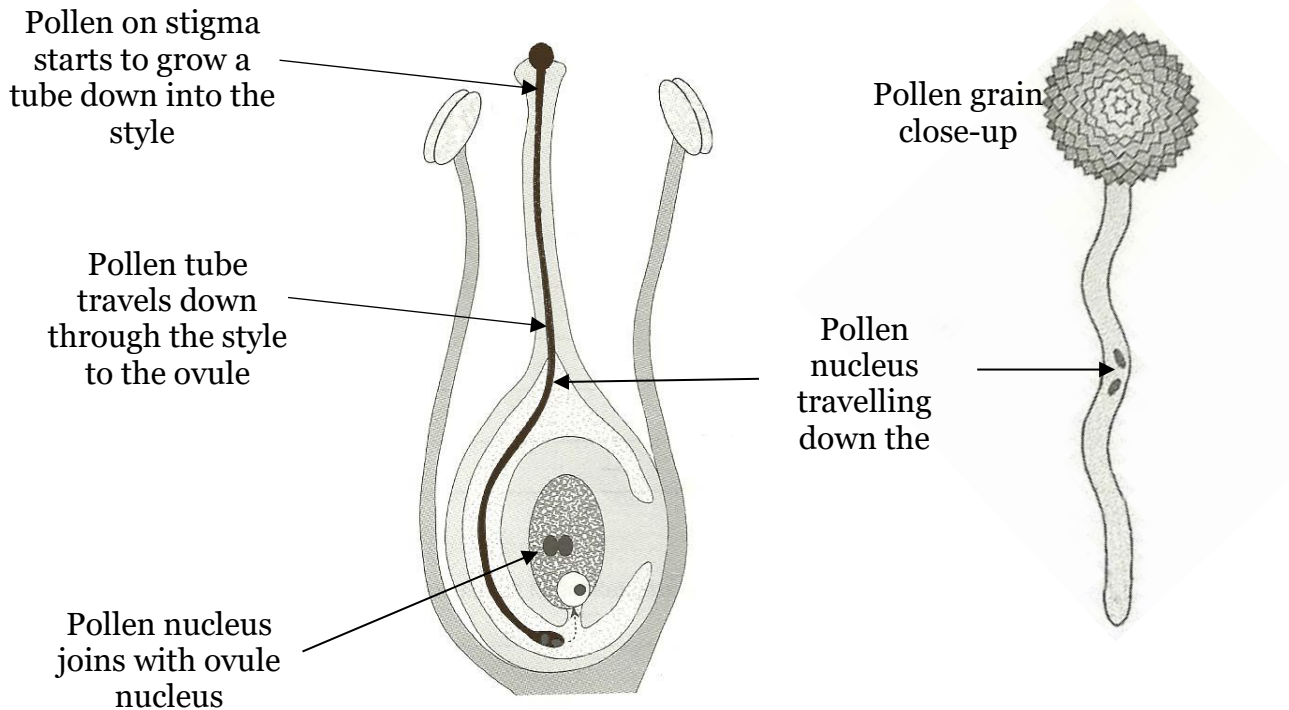


Task 4: Complete the table below:

Insect pollinated	Wind pollinated
	Petals small and dull
Anthers inside the flower	
Nectary present	

Fertilisation

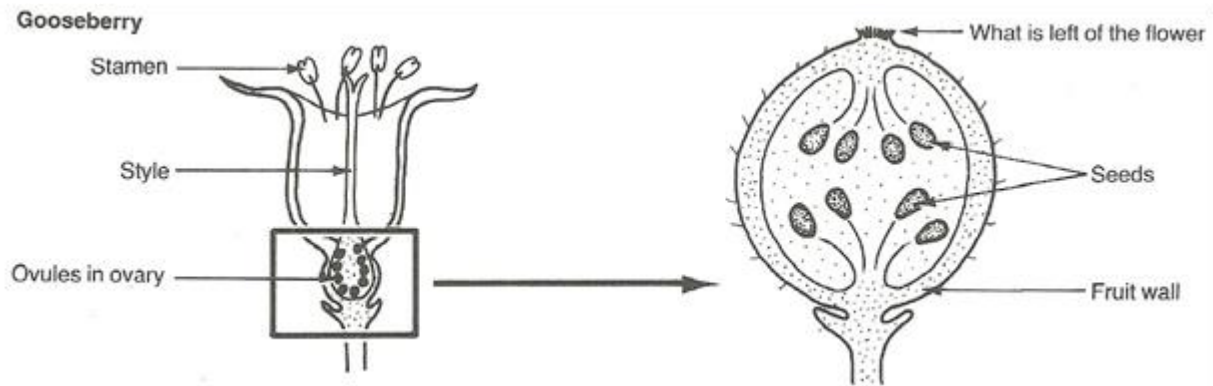
After pollination, the **nucleus** from the pollen grain must travel from the stigma down to the ovule to achieve fertilisation. **This is when the nucleus of the pollen joins with the nucleus of the ovule.**



Questions

1. A pollen grain attaches to the stigma of a flower – this is called _____ .
2. The pollen grain then grows a _____ down from the stigma into the style and then further down to the _____ inside the _____ .
3. The pollen _____ then travels down the tube to reach the _____ .
4. The nucleus of the pollen grain joins with the _____ of the _____ . This process is called _____ .

Fruit formation



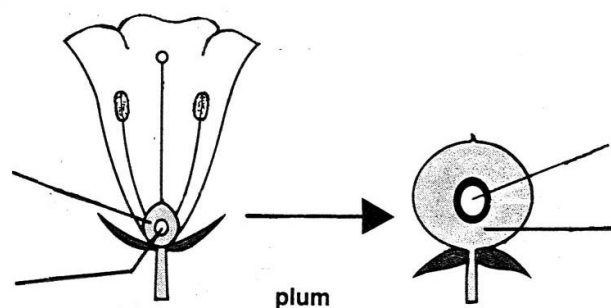
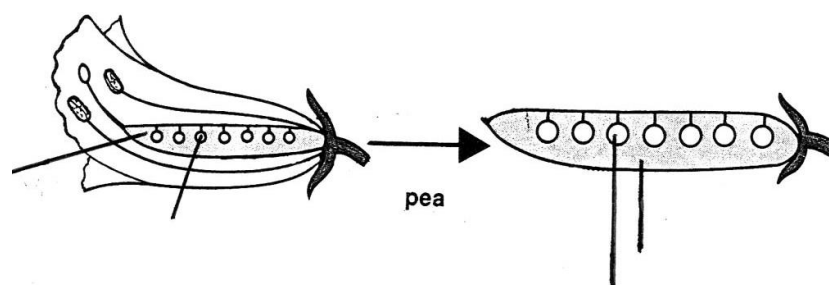
Questions

1. Which part of the flower develops into the **seed** after fertilisation?

2. Which part of the flower develops into the **fruit wall** after fertilization

3. What happens to the other parts of the flower after fertilisation?

4. Label the following structures on each of the diagrams below:
• fruit wall • ovary • ovule • seed.



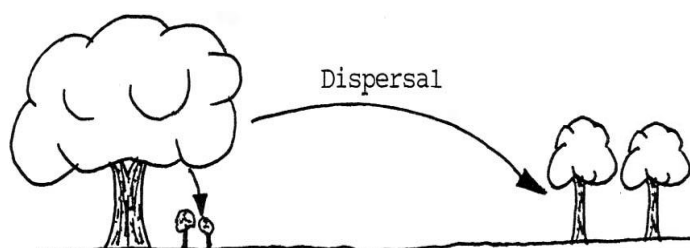
5. Use blue to colour the **ovary** and the structure that it forms in the fruit.
6. Use yellow to colour the **ovule** and the structure that it forms in the fruit.
7. How many ovules would you expect to find in these flowers? (Rough guesses for some!):

Apple =	Avocado =	Tomato =
Peach =	Kiwi fruit =	Cherry =

Seed dispersal

The function of the fruits is to disperse the seeds **away from the parent plant**.

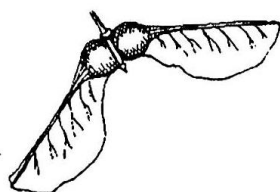
This dispersal means that the new seedlings will have a **better chance to grow**.



1. Why will the dispersed seedlings be able to grow better?

Seed dispersal methods: how are the following seeds adapted for dispersal?

Wind dispersal



Sycamore

Adaptations of **second** type of fruit:

Animal-internal dispersal



Blackberry (Bramble)

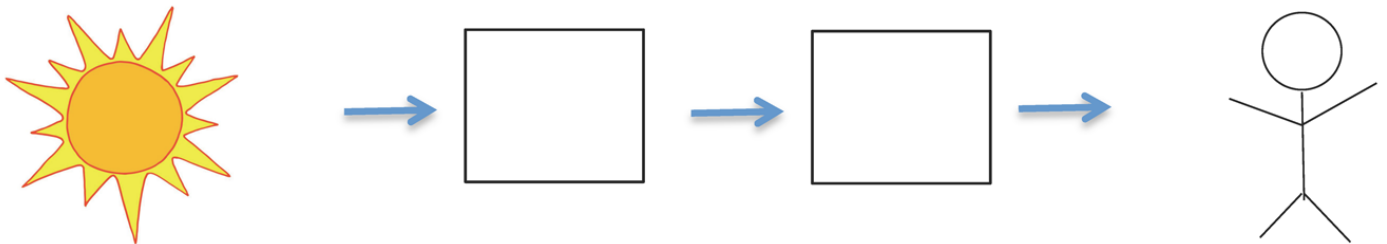
Adaptations of **fruit** to help dispersal:

Lesson 6 – Producers and Consumers

Food from plant into animal

Plants are **producers** – they are the only living things that can make their own food. Animals and fungi are **consumers** because they can only get their food by eating other living things.

On the diagram below, draw and label an appropriate organism in each box



Task 5: We use different parts of plants as foods. Identify the parts of the plant we eat in the following plants:

oranges



spinach



peas



celery



apples



lettuces



artichokes



asparagus



corn

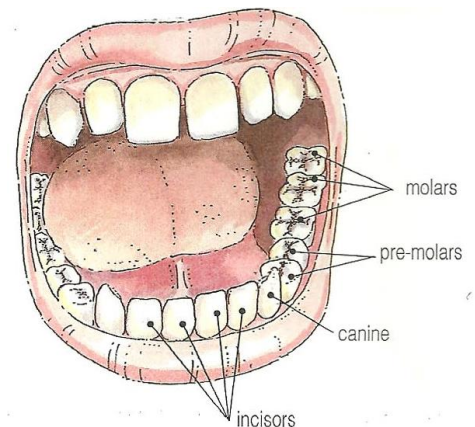


We have learned that plants **make food (sugar) for themselves** and that they store it as starch. This storage can be used by animals that **feed on** plant parts.

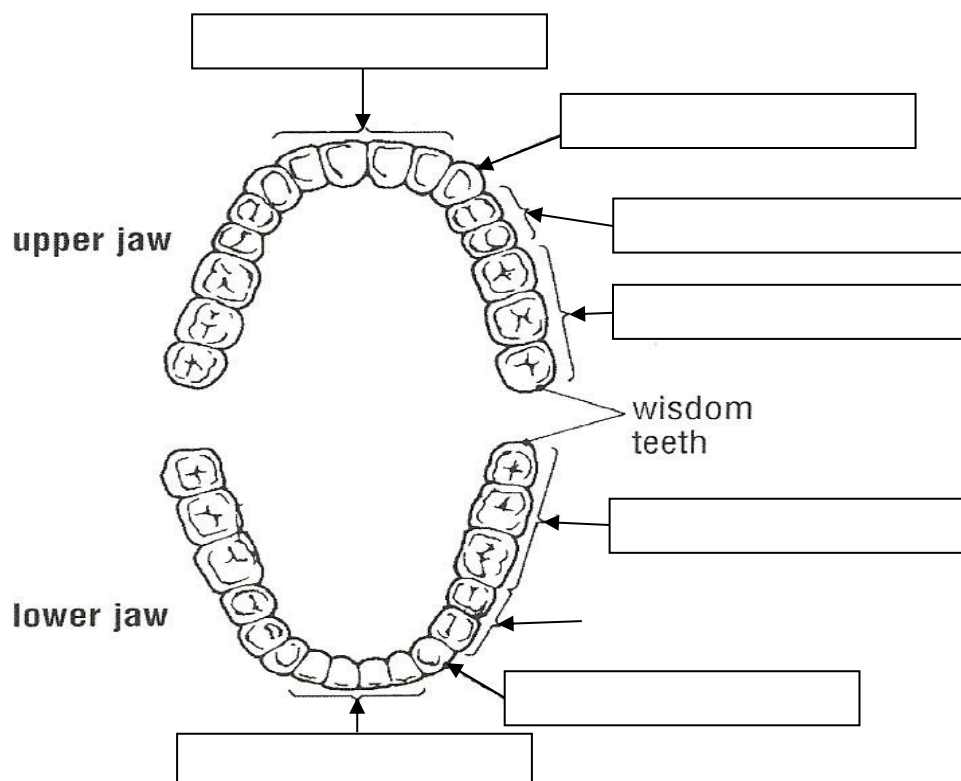
Animals **bite and digest** the plant parts and use the energy released for growth and movement. They are **consumers**.

Tooth truths

Teeth do the hard work at the start of the digestive system. They break food up into smaller chunks so that you can swallow it.



Task 6: Label the diagram below to show the four different tooth types:



Use colour key below and **lightly shade** the four different tooth types on the diagram.

Incisor	Canine	Pre-molar	Molar

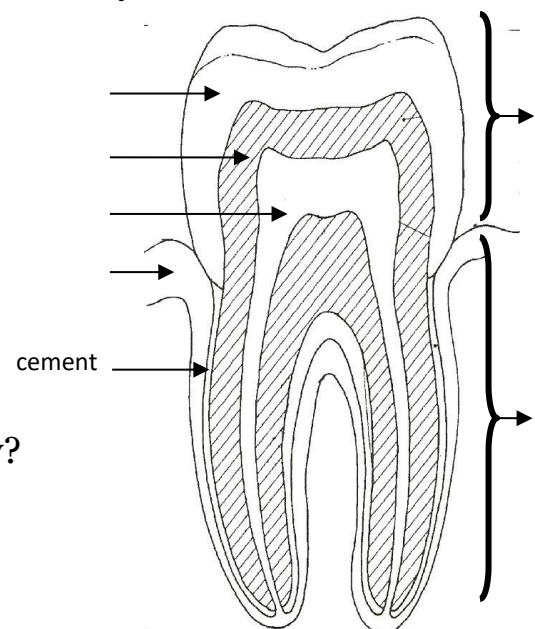
Not all animals rely on plants directly for energy, some animals hunt and kill plant eating organisms to get their energy.

Activity 3: Skulls and teeth jigsaw. In groups, match the organism to its food type and identify the structures on their skull that are suited to this food type?

	Herbivore	Carnivore	Omnivore
Name of organism			
Type of teeth			
Food eaten			
Skull shape and special structures			

Task 7: Label the following structures on the cutaway human tooth.

- crown
- enamel
- dentine
- gum
- root
- pulp cavity



Which part is the hardest material in the body?

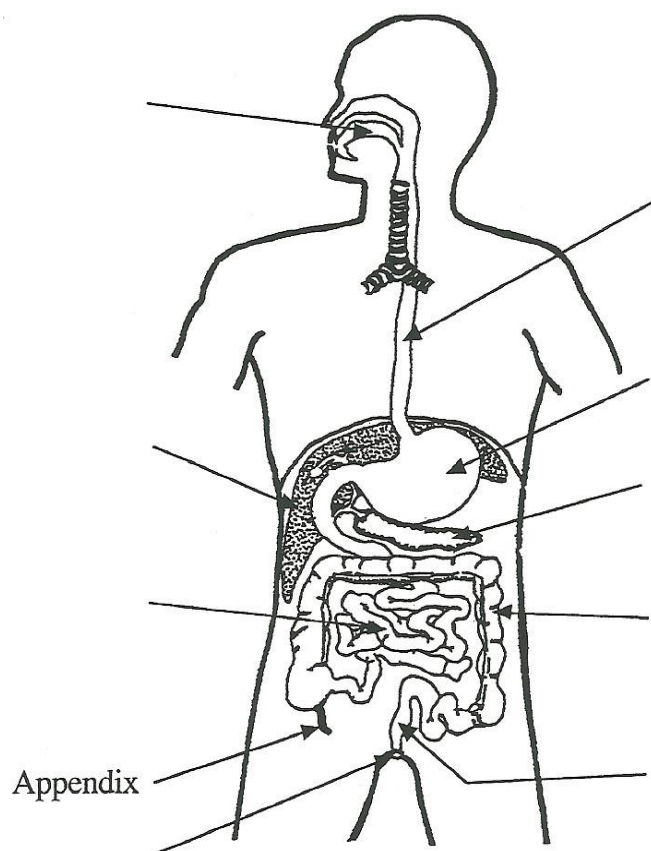
Which area of the tooth has nerves and blood vessels?

Add some colours to the diagram to make the different areas clear.

Lesson 7 – Breaking down food

Task 8: Label and colour the following parts on the diagram below:

- anus
- oesophagus (pink)
- large intestine (brown)
- liver (purple)
- mouth (red)
- pancreas (yellow)
- rectum (brown)
- small intestine (pink)
- stomach (pale brown)



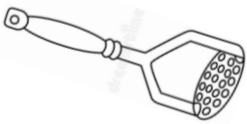
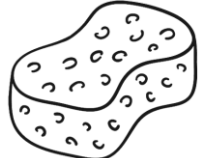
Write down the correct order of the parts that food **passes through** in the digestive system.

Mouth

1. _____
2. _____
3. _____
4. _____
5. _____
6. _____

... and then the undigested remains pass out through the **anus** (at an appropriate time!).

Activity 4: As you watch the demonstration of ‘GUM to BUM’ led by your teacher, complete the table to describe what each item **represents** and/or its **role** in a real digestive system.

<i>Item</i>	<i>Represents</i>	<i>Role in digestive system</i>
		
	Enzymes	
		Makes ideal conditions for enzymes to break down proteins
		1. 2.
	Bile salts	
		
	Blood stream	1. 2.

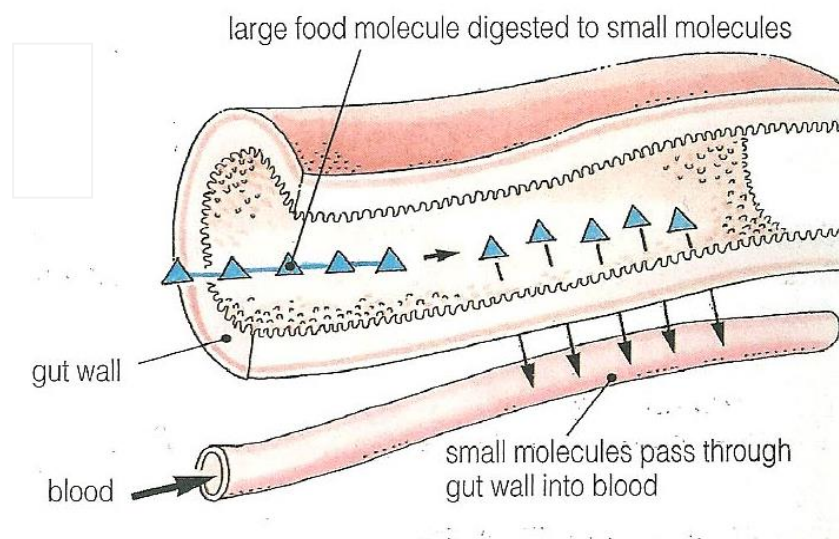
Use this list to complete the table below.

- bile salts
- stomach acid
- teeth
- enzymes
- stomach muscles

Helps to break down the food <i>physically</i> into smaller chunks.	Helps to break down food molecules <i>chemically</i> to form smaller molecules.
--	--



The diagram below shows the working of the guts.



3. Label the diagram with the following names:

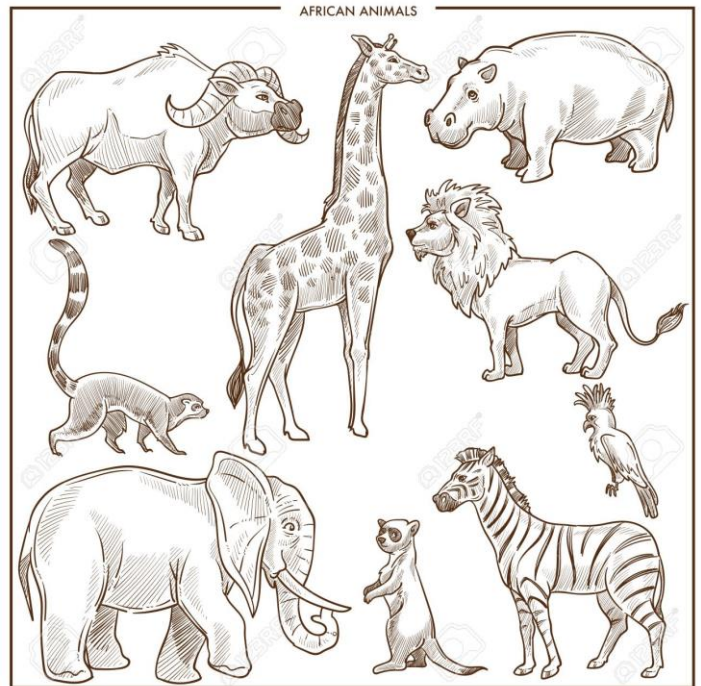
- sugar molecules
- starch molecules

4. Why can starch molecules not get into the blood?

Optional – Adaptations to a plant diet

Not all herbivores feed the same way, let's look at different types of mammals.

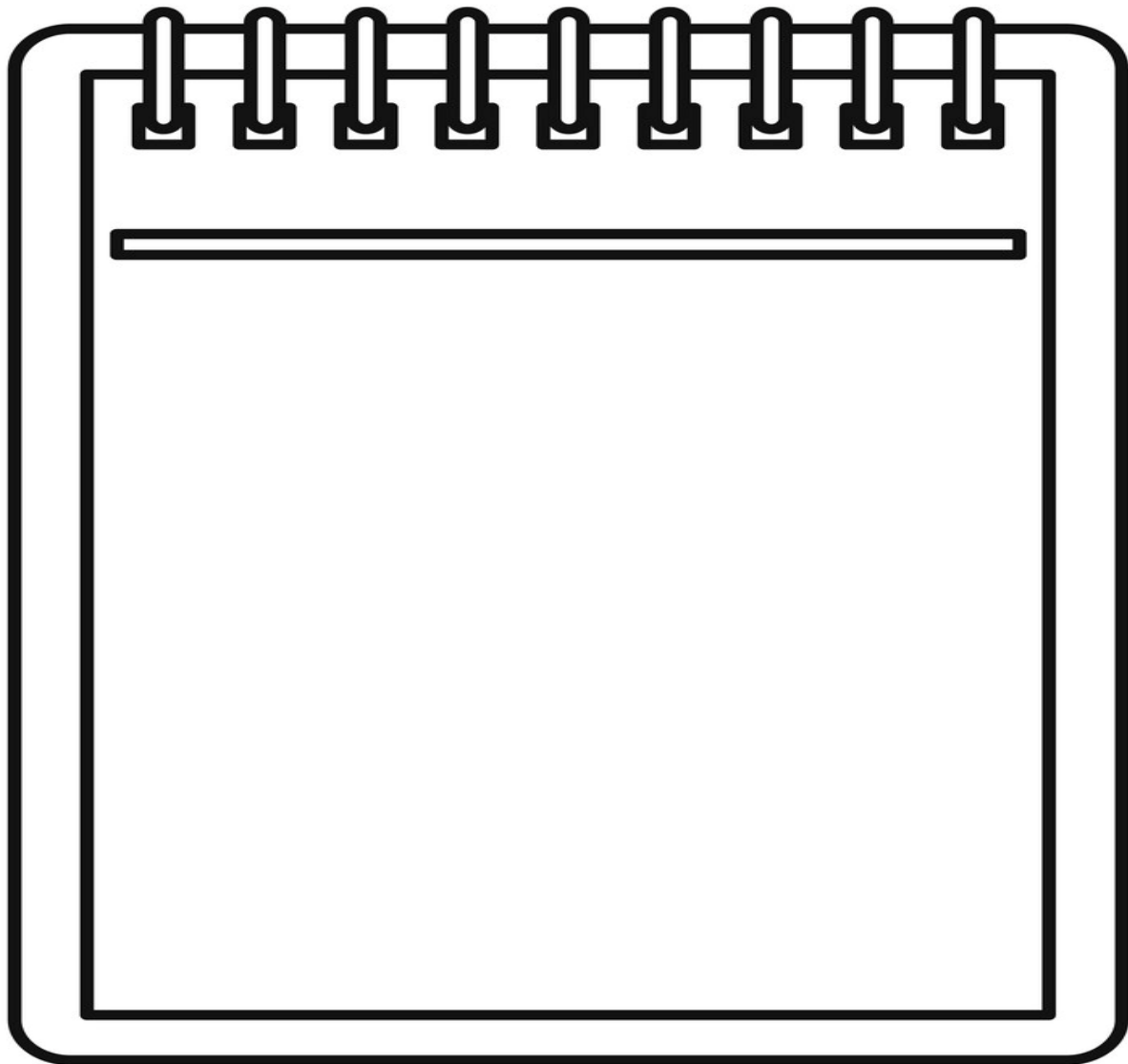
The purpose of the digestive system is to break food into particles small enough to pass through the gut to enter the bloodstream, which transports these nutrients to all the cells in the body. Yet, depending on an animal's feeding behaviour and diet, regions of the gut used for the transport, storage, and processing of food have evolved specializations.



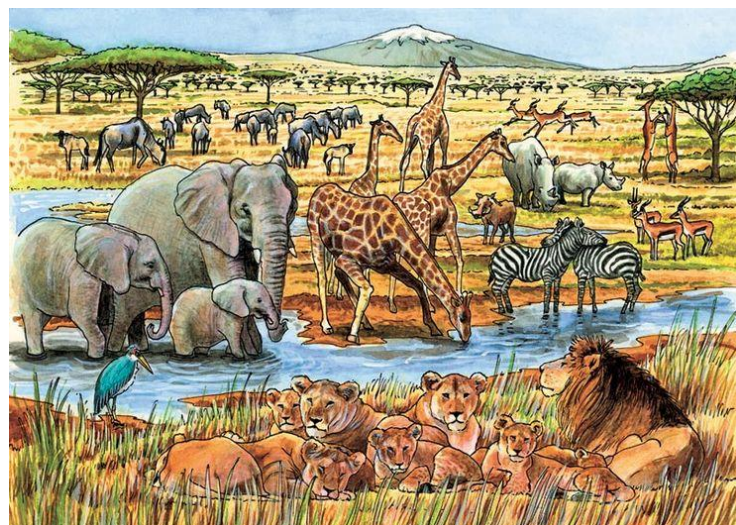
This research task compares the digestive systems of two herbivores feeding together on the fibrous, nutrient-poor grasses of the African savanna: a buffalo and a zebra. Because grass contains less usable energy than other plant parts such as fruits or nuts, these grazers need to spend a relatively large portion of their day chewing and digesting food to obtain sufficient energy.

To reach the nutrients available in the body of a plant cell, the cellulose-rich cell wall must be broken. No mammal has the necessary chemicals to **digest cellulose** to capture the energy stored in this complex sugar. Instead, the mammalian digestive tract harbors microbes that perform the digestive work, releasing simpler, absorbable molecules.

Activity 8. 1. You will compare the strategies of ruminant and non-ruminant herbivores using the information provided by your teacher and summarise your findings here:

A large, empty spiral-bound notebook with a black outline and a horizontal line near the top, intended for summarizing findings.

If zebras, rhinos, and antelopes all eat grass, why don't they fight over food on the African savanna?



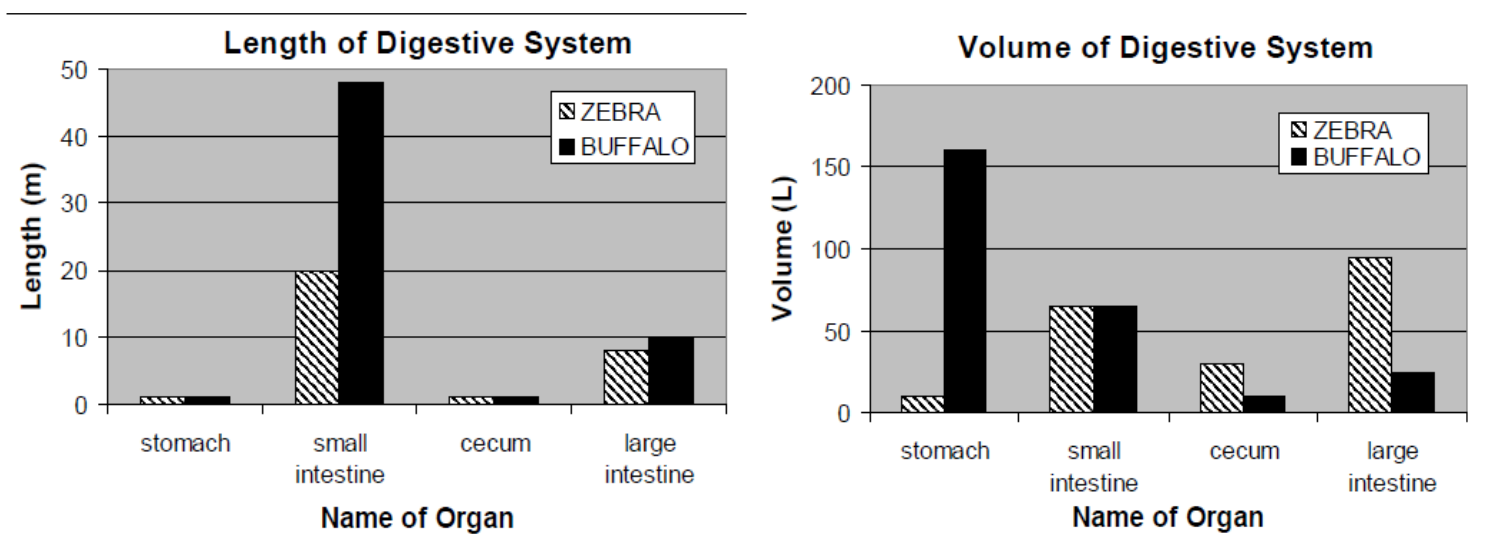
Non-ruminants like zebras will eat the lower-quality tips of the grass, which is older and more fibrous. It will pass through their system quick enough. Then gazelles, as ruminants, can eat the younger, low-cellulose shoots that are left exposed. They won't have to eat nearly as much but will want their rumen to take advantage of all the nutrients in the plant. **So, each hoofed animal fills their own niche, or environmental role.**

As the rumen bacteria digest cellulose, gas is produced. **When the bacteria in our small intestine digest vegetables, we also produce gas**, and expel it by farting. However, the ruminant digestive tract is so long, that it would be uncomfortable to expel gas out the back, so instead, they belch all day! How does this byproduct contribute to global warming?

If you were wandering amid a mixed herd of zebras and buffalo, you might be able to tell apart their dung. How? The zebra, who can't digest plants completely, will have dung with coarser, undigested grass sticking out. The dung of the buffalo, a ruminant, should be more uniform throughout.

Activity 2: data handling

Compare the digestive systems of a buffalo and a zebra.



Fill in the table reading data from the graphs:

	Ruminant	Non-Ruminant	Ruminant	Non-Ruminant
	Length		Volume	
Stomach				
Small intestine				
Cecum				
Large intestine				
TOTALS				

Questions:

1. Why does the buffalo's stomach hold so much food?
2. Why does the small intestine need to be so long?
3. How is the zebra stomach different and what does that mean for the animal?
4. How might the plant matter in the large intestine of the zebra be different to that of the buffalo?



Optional research task: What is the problem with PALM OIL TREES?

We have learned that plants are good, they provide us with all kinds of benefits, so surely planting any type of tree is good? Look into the issue with large areas of land dedicated to Palm oil trees OR avocado farms OR any "monoculture" of forest.

Practice Questions

Lesson 1 Questions – Life giving plants

1. State 4 resources humans depend on that plants provide:

2. Without plants, what negative effects would be seen in the environment?

3. What type of plant has tubes that funnel water to the top of the plant?

Lesson 2 Questions – Parts of a Plant

4. What are the functions of the following parts of a plant;
 - i) leaves

 - ii) stem

 - iii) flowers

 - iv) roots

Lesson 3 Questions

5. What process do plants carry out to produce sugars?

6. What materials do plants need to carry this process out?

7. What do plants store their sugar as in their plant body?

8. What is the positive test for this substance in the lab?

Chemical needed for test:

Colour of Positive result:

Lesson 4 questions – Flower Power

9. Where is pollen made?

10. What is the role of petals?

11. What 4 structures make up the carpel?

Lesson 5 questions – Pollination

12. What does the term pollination mean?

13. Describe the journey of pollen nucleus, from anther to ovary.

14. What needs to happen for the ovary to grow into a fruit?

Lesson 6 questions – Producers and Consumers

15. Add examples to the table below of food we eat for each part of the plant:

Stem	leaves	roots	Flower	fruit

16. For each statement below, decide if it is true or false, then correct the underlined word if you identified any as false.

statement	True or false	correction
An <u>herbivore</u> eats both plants and animal material to gain energy		
When eating an artichoke, you eat the <u>flower bud</u>		
The 4 types of teeth are <u>incisor</u> , <u>canine</u> , <u>post-molar</u> and <u>molar</u>		

Lesson 7 Questions – Breaking down food

17. Write the order of food as it passes through the digestive system as a flow diagram.

18. Why do large insoluble molecules need to be broken down chemically?